

Using LabView to Explore Symmetry Breaking in a Coupled Logistic Map Circuit

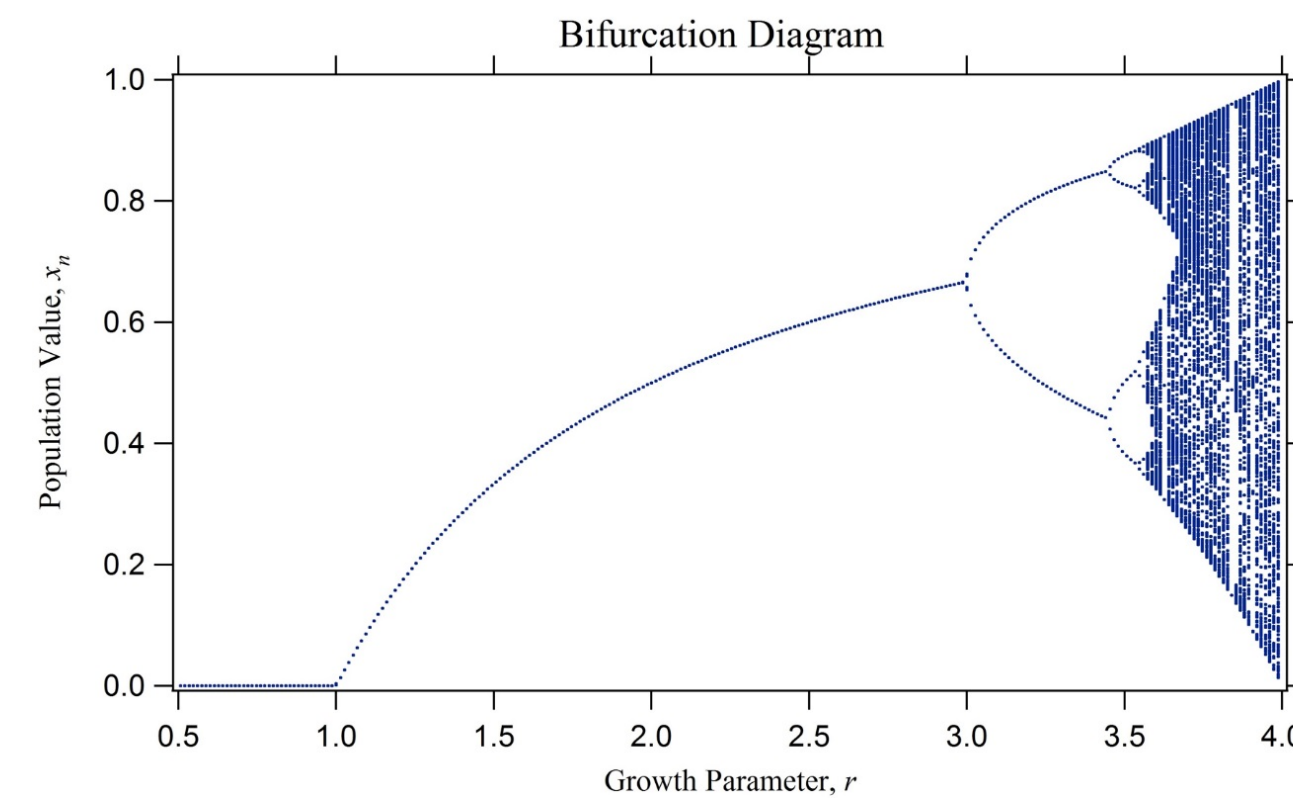
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Coupled Logistic Map Model

$$x_{n+1} = rx_n(1 - x_n)$$

The **Logistic Map** is a demographic model that describes the evolution of a population. x_n (between 0 and 1) represents the ratio of current population and the maximum capacity, and r (between 0 and 4) is the growth rate.

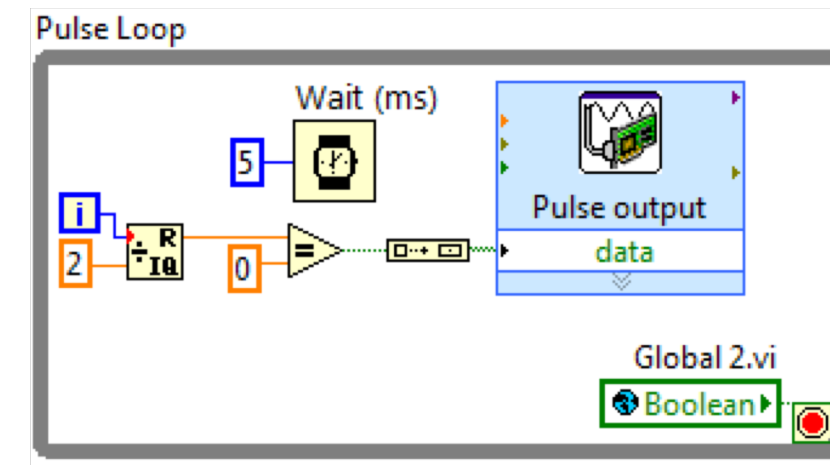
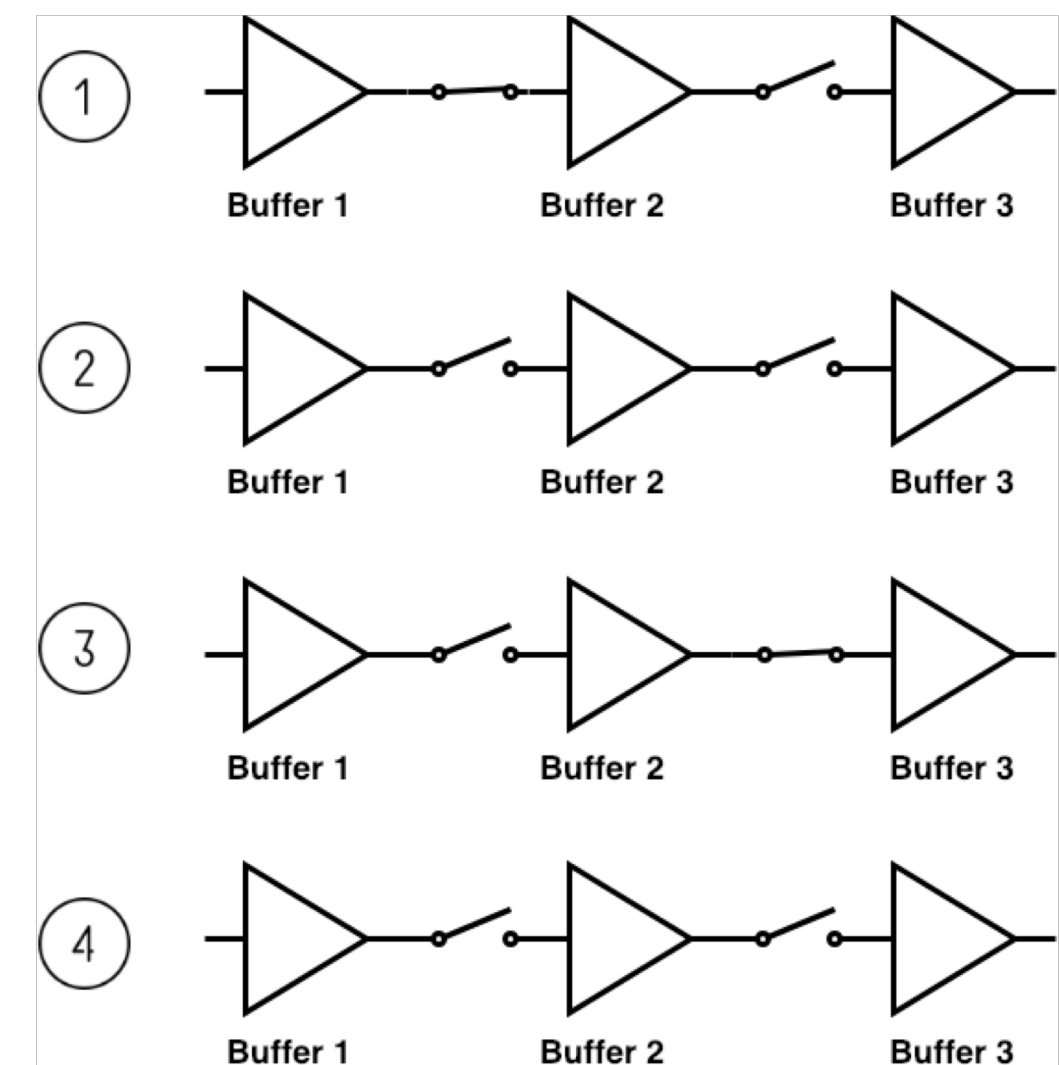


$$\begin{aligned} x_{n+1} &= (1 - \epsilon)r_x x_n(1 - x_n) + \epsilon r_y y_n(1 - y_n) \\ y_{n+1} &= (1 - \epsilon)r_y y_n(1 - y_n) + \epsilon r_x x_n(1 - x_n) \end{aligned}$$

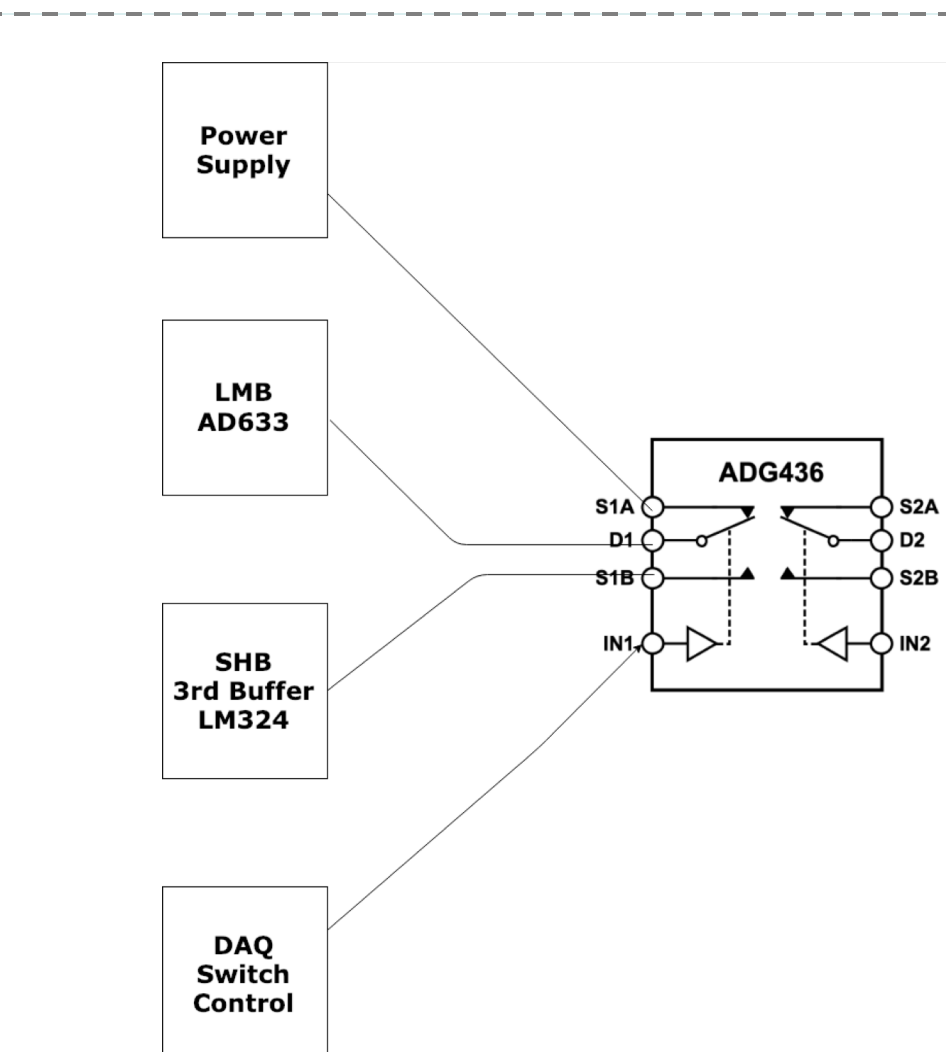
The **coupled Logistic Map** adds in the interactive effects between populations of two species that live in the same environment. x_n and y_n stand for the ratios of species x and y respectively, r_x and r_y are their growth rates. ϵ is a new parameter between 0 and 1 that indicates the coupling strength between the two populations.

Experimental Setup

The coupled logistic map circuit consists of three parts: the logistic map block (LMB), the sample and hold block (SHB) and the coupling block (CB). SHB discretizes the processing of the circuit so that we can realize the discrete model, LMB acts as the logistic map function which takes the previous voltage value and produces a new voltage value, and CB adds in the influence from the other LMB output [2].

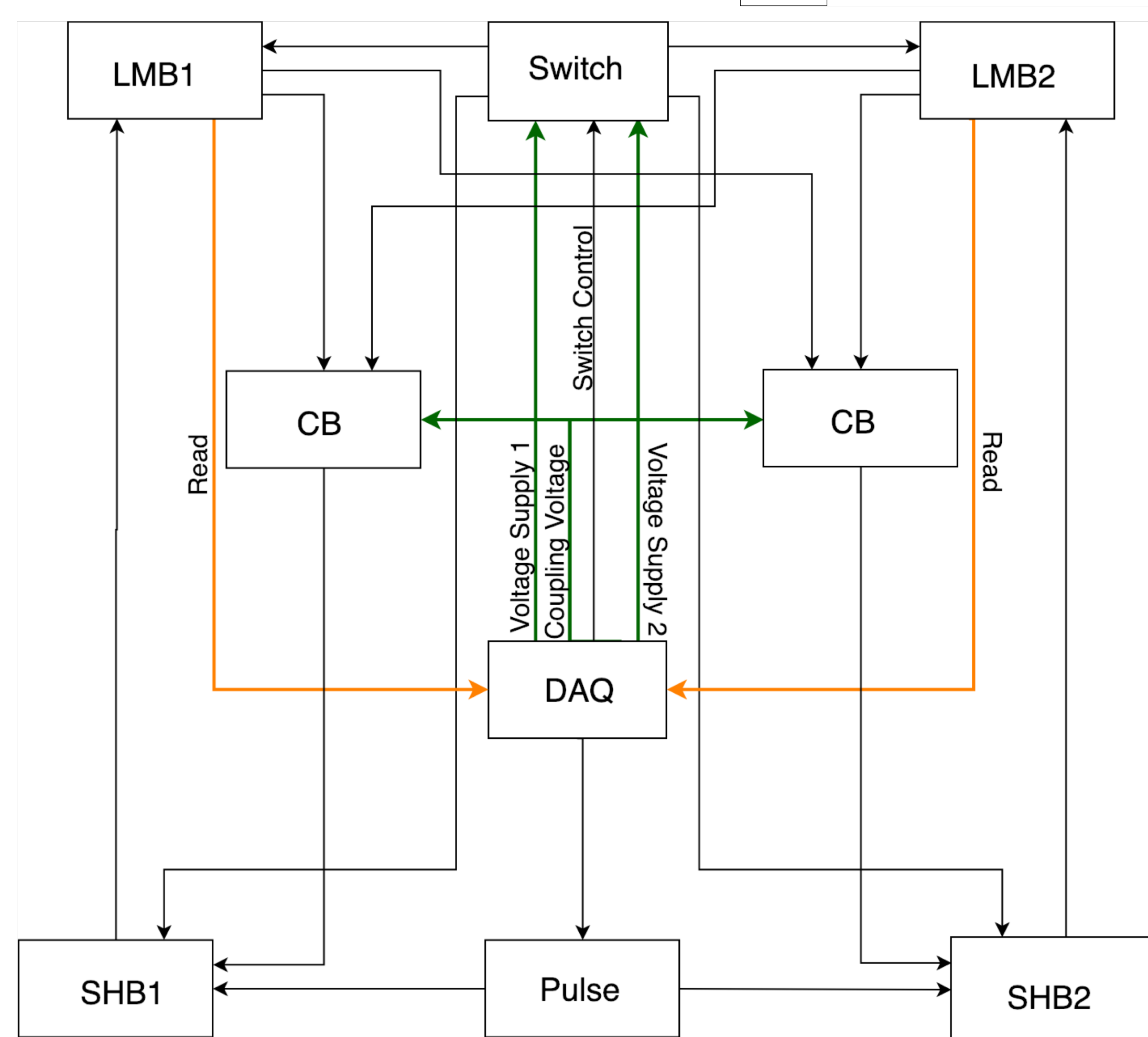


The signals to the switches are controlled by LabView.



Buffers in SHB are triggered so that a new value gets produced with one half of the frequency of the clock.

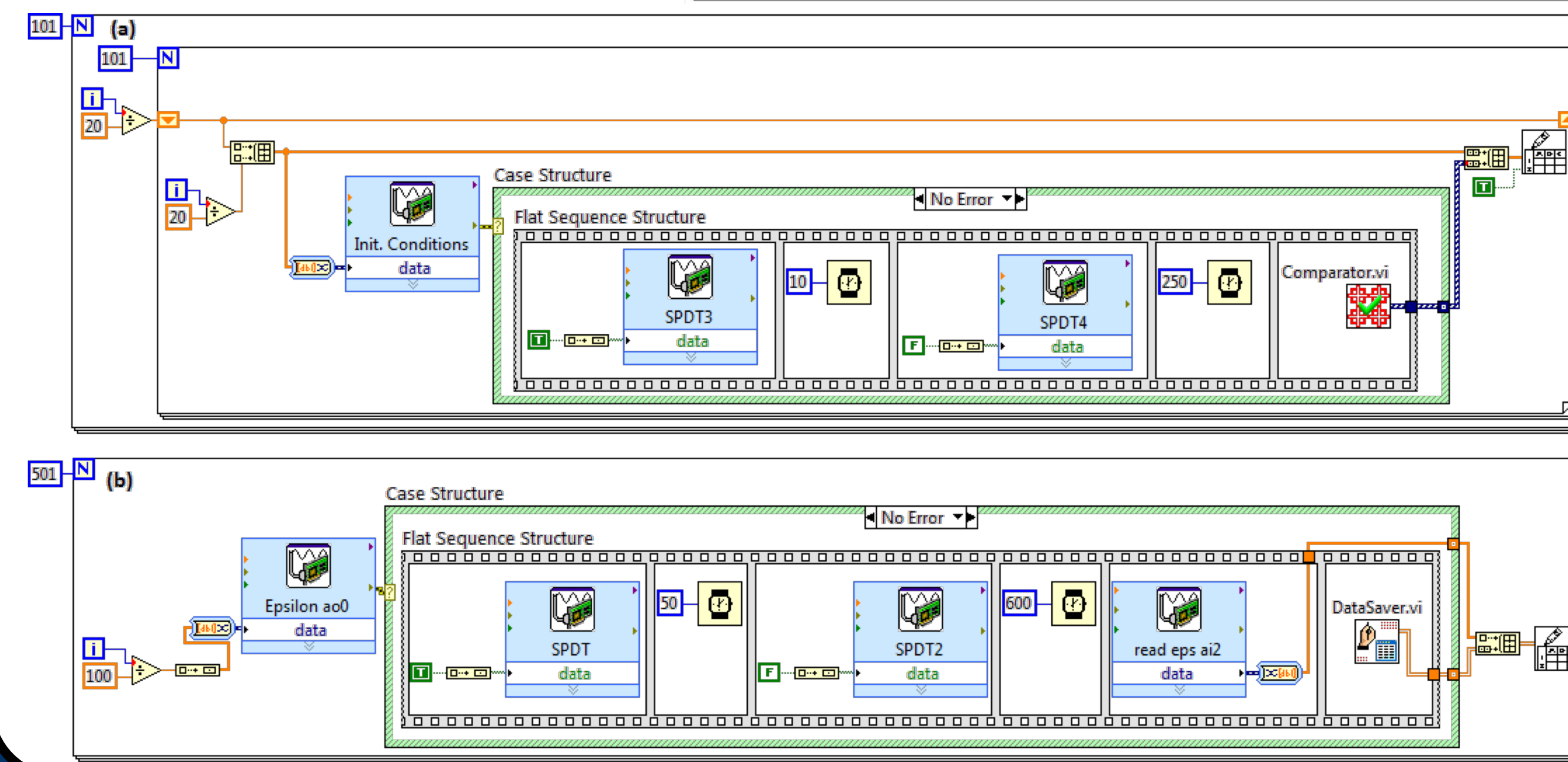
We used a SPDT analog switch (ADG436) to send in the initial conditions for the two logistic maps.



Data Acquisition (LabView)

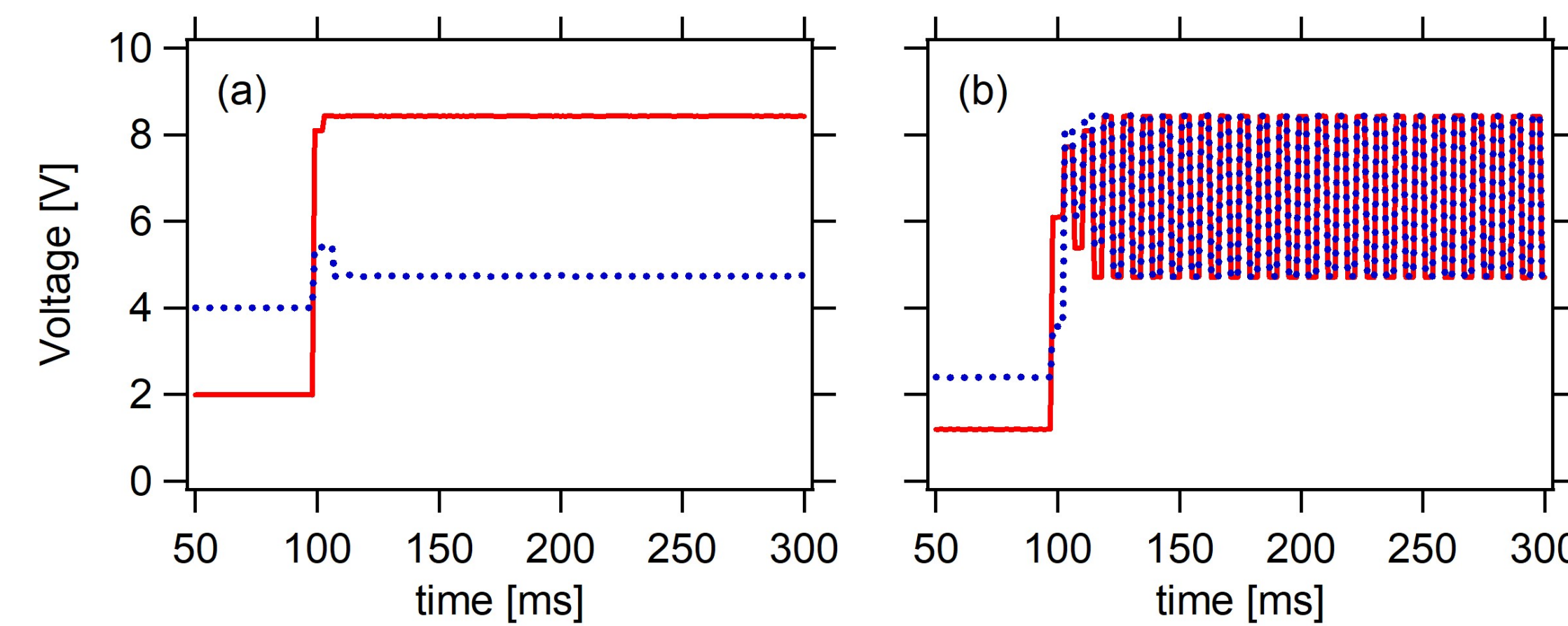
We used LabView to send in digital control to the switch and the initial voltages to the two LMBs. LabView can be used to interface with the DAQ, read in the time-series and produce waveform graphs with given initial conditions.

- (a) LabView code for basins of attraction measurement
- (b) Making a bifurcation diagram of ϵ

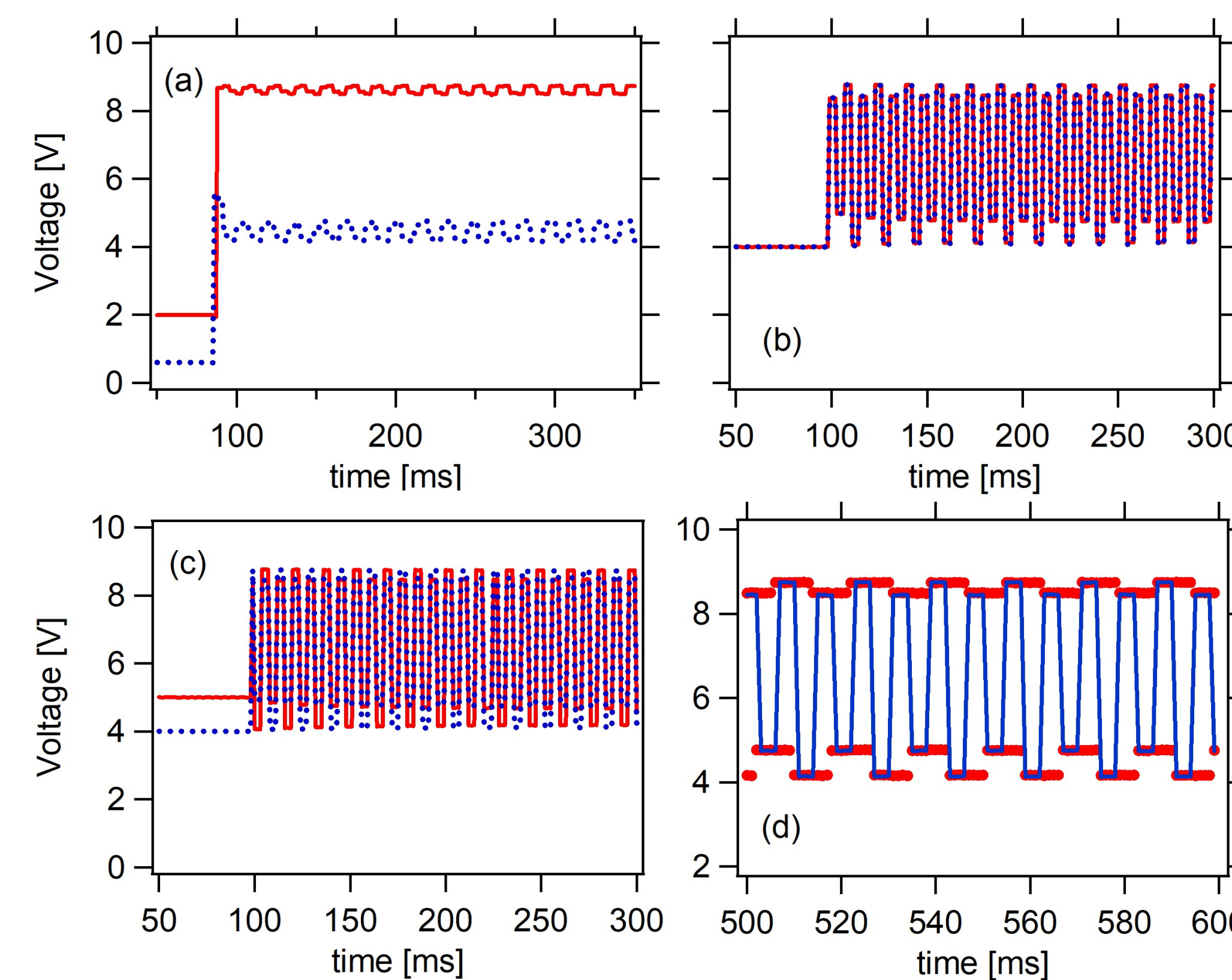


Results

Time-series with $\epsilon = 1$

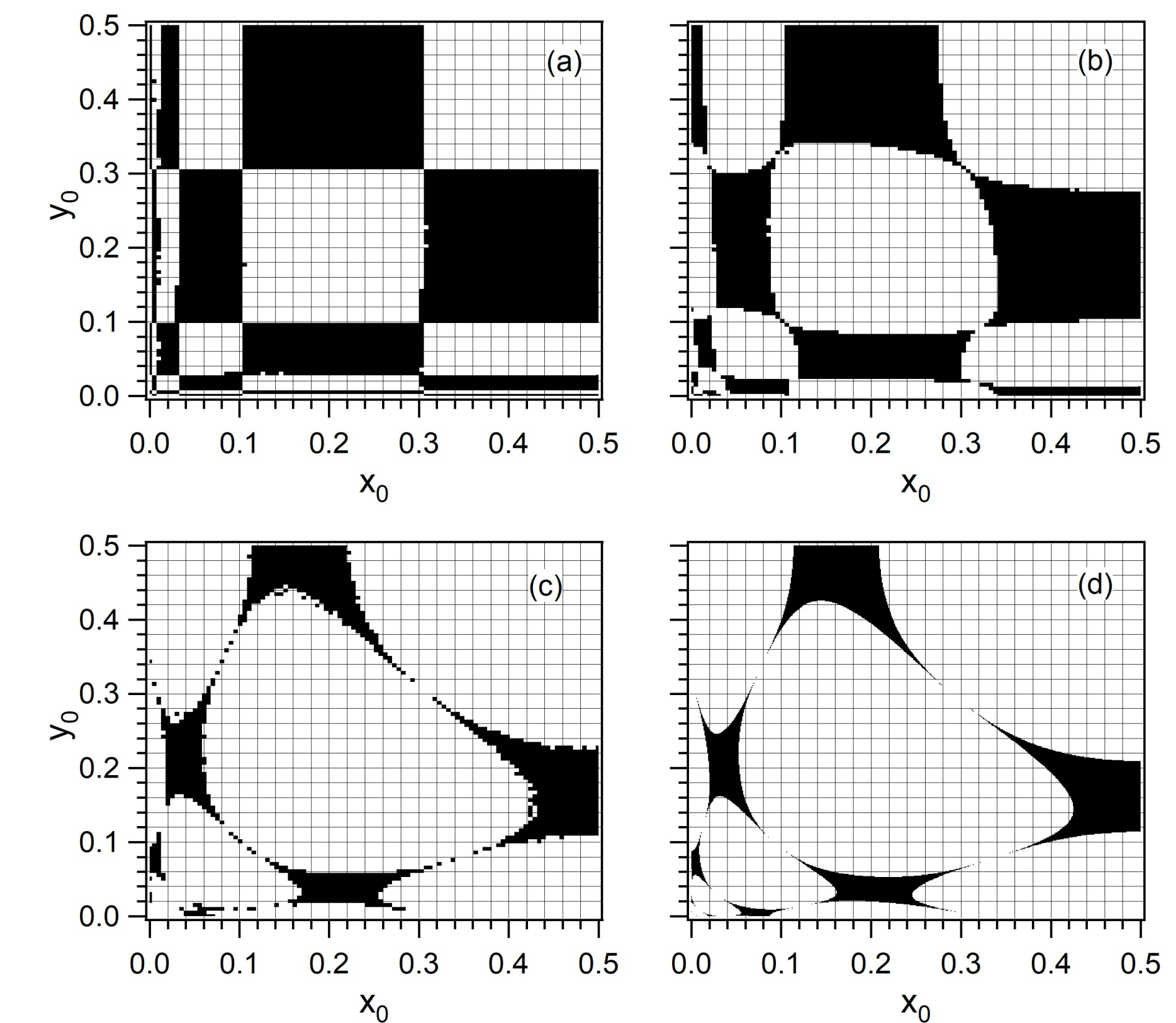


Symmetric and symmetry-broken states for $r = 3.4$. (a) with initial condition (2V, 4V) and (b) with (1.2V, 2.4V)



$r = 3.5$. (a) symmetry-broken state, (b) symmetric state, (c) phase-shifted state, and (d) shows the period of the symmetry-broken solution is double of the period of the symmetric solution

Basins of Attraction

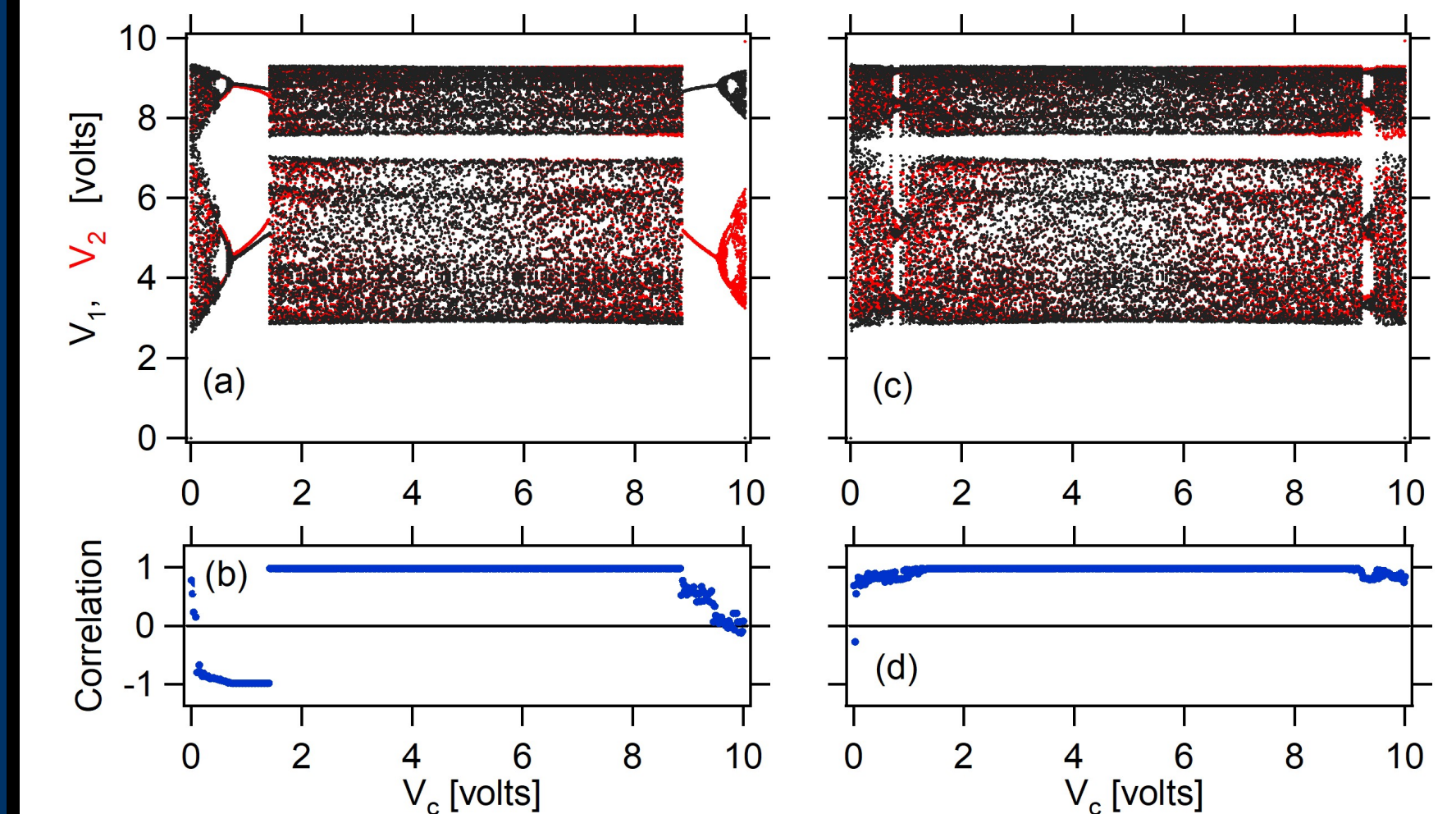


Set a fixed ϵ each time and keep $r = 3.4$. The initial conditions for x and y are scanned through from 0 to 0.5.

Black $\rightarrow$ Symmetry-broken
White $\rightarrow$ Symmetric

- (a) $\epsilon = 1$
- (b) $\epsilon = 0.95$
- (c) $\epsilon = 0.9$
- (d) numerical simulation for $\epsilon = 0.9$

Bifurcation Diagram of ϵ in Chaotic Regime



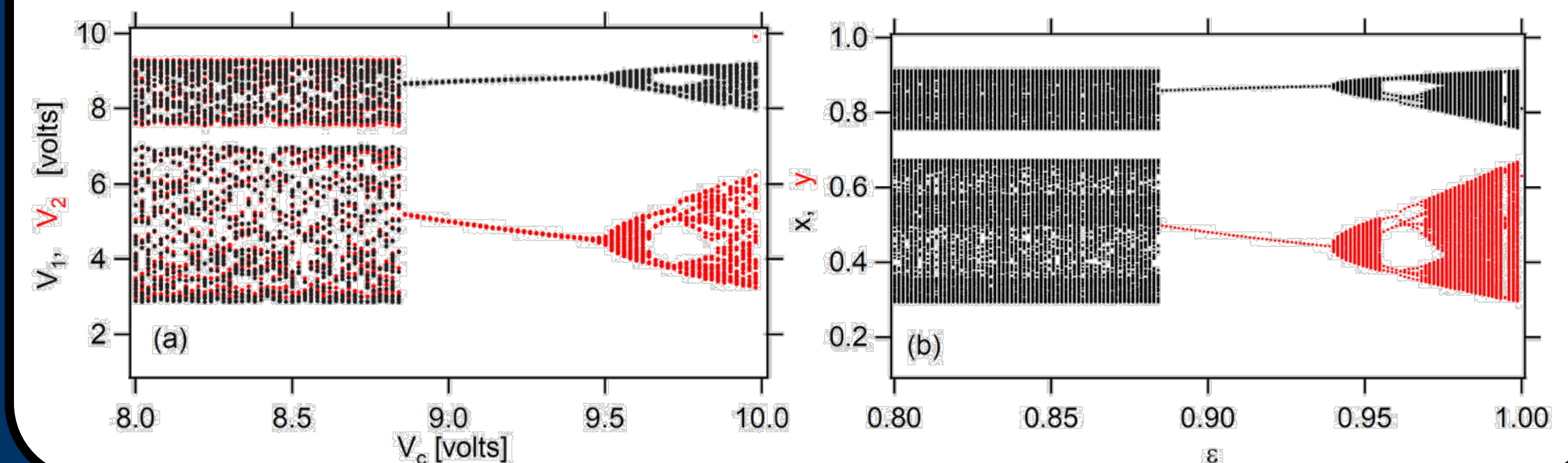
Pearson correlation coefficient

$$\rho_{xy} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{(N - 1)\sigma_x \sigma_y}$$

- $r = 3.65$
- (a) initial condition (0.5, 0.2),
- (b) initial condition (0.5, 0.5)

A zoomed-in picture of the bifurcation diagram illustrating the transition from synchronized chaos to symmetry-broken phases.

(a) Experimental data, (b) numerical results



Conclusions

Initial conditions of coupled logistic maps can be sent in by adding a switch control. By implementing LabView, we are able to physically observe the symmetric, the symmetry-broken and the phase-shifted states of a coupled logistic map circuit with given growth rates, coupling strength and initial conditions. The ability of tweaking different parameters allows us to explore how they each affect the behavior of the map. Basins of attraction with given ϵ and r , and bifurcation diagram of ϵ can be experimentally produced using the setup. From the bifurcation diagram, we observe a transition from synchronized chaos to symmetry-broken chaos. The theoretical foundation of this nature can be further investigated. In the future, we can explore a coupled ring of logistic maps by extending the number of components in the circuit.

REFERENCES

- [1] Han, J., Jones, S., & Skowronski, K. (2017). "Exploring the Correlated Dynamics of Coupled Logistic Map Systems", Thesis, Dickinson College.
- [2] L'Her, A., Amil, P., Rubido, N., Marti, A. C., & Cabeza, C. (2016). "Electronically-implemented coupled logistic maps", *European Physical Journal B* **89**, 81.
- [3] Wynne, E. (2018). "Using an Arduino in a Coupled Logistic Map Circuit to record Bifurcations and Chaos", Thesis, Dickinson College.